

DRUG REPURPOSING DEMO

Query : 'psoriasis' → psoriasis

Model : HGT

Scoring 100 drug candidates ...

Computing disease similarity ...

OUTPUT 1 — TOP 5 CANDIDATE DRUGS

Disease : psoriasis | Model : HGT

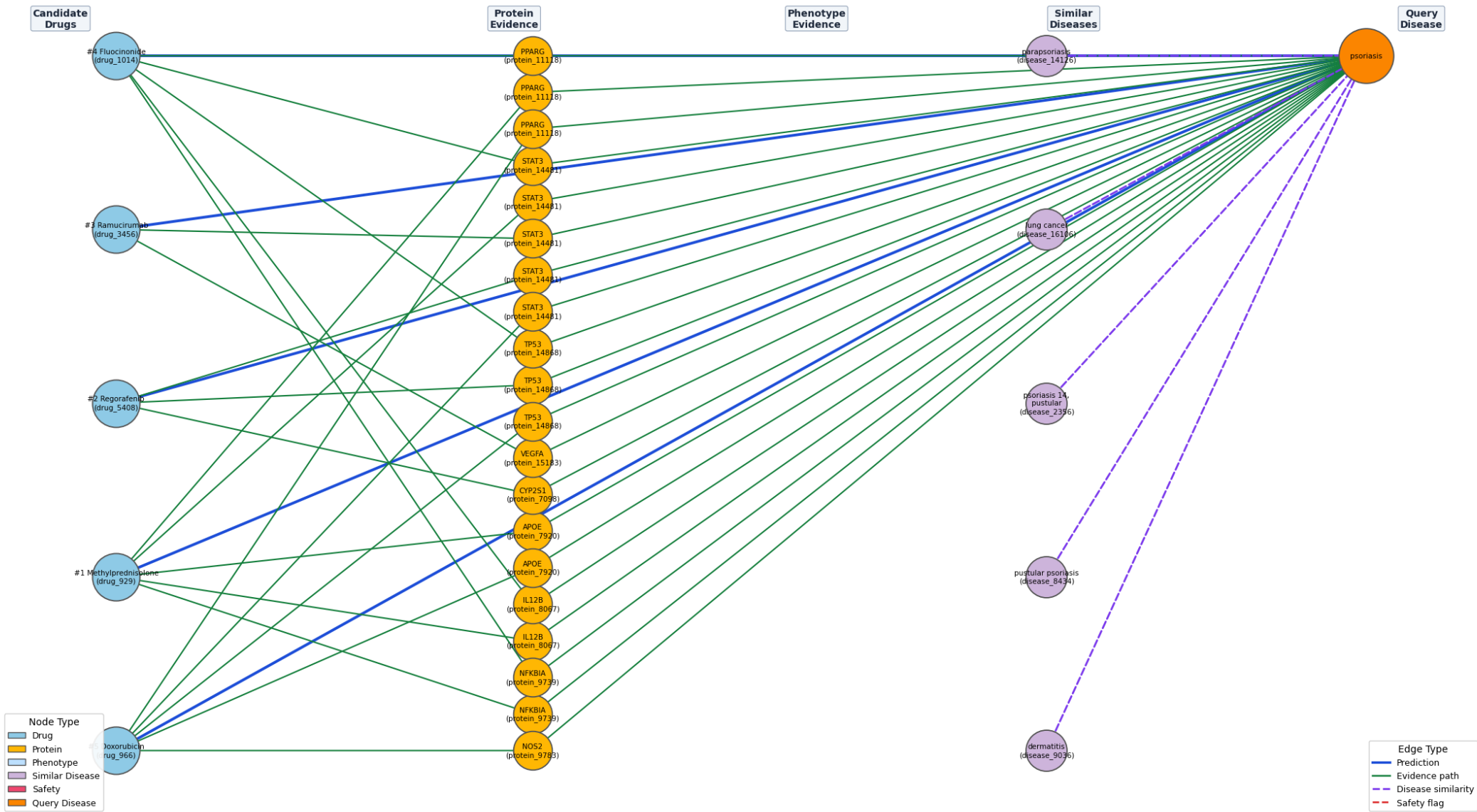
Rank	Drug	Model Score	Sim. Support	Rerank Score	Risk	Evidence	Safety Flags	Known Treatment For
1	Methylprednisolone (drug_929)	0.573084	0.673471	0.683564	low	6	0	X-linked adrenal hypoplasia congenita, chondrocalcinosis, dermatitis, atopic (+73 more)
2	Regorafenib (drug_5408)	0.572930	0.565849	0.656573	low	5	0	gastrointestinal stromal tumor, pediatric hepatocellular carcinoma, adenocarcinoma of liver and intrahepatic biliary...
3	Ramucirumab (drug_3456)	0.572884	0.851147	0.627873	low	2	0	pediatric hepatocellular carcinoma, undifferentiated carcinoma of liver and intrahepatic biliary tract, liver cancer...
4	Fluocinonide (drug_1014)	0.572963	0.326529	0.596762	low	6	0	primary cutaneous T-cell lymphoma, discoid lupus erythematosus, dermatitis (+8 more)
5	Doxorubicin (drug_966)	0.572843	0.285298	0.586388	low	6	0	acute myeloid leukemia and myelodysplastic syndromes related to topoisomerase type 2 inhibitor, acute myeloid leukem...

OUTPUT 2 — TOP 10 SIMILAR DISEASES

Rank	Similar Disease	Similarity Score	Embedding Cos	Shared Proteins	Shared Effects	Shared Drugs
1	parapsoriasis (disease_14126)	0.794633	0.984647	59	0	7
2	psoriasis 14, pustular (disease_2356)	0.721991	0.849412	57	0	0
3	dermatitis (disease_9036)	0.666350	0.895644	2	0	4
4	lung cancer (disease_16106)	0.582209	0.954773	14	0	1
5	pustular psoriasis (disease_8434)	0.581464	0.986316	1	0	0
6	rheumatoid arthritis (disease_15665)	0.579723	0.979051	14	0	3
7	hepatocellular carcinoma (disease_14718)	0.577839	0.976508	17	0	0
8	liver cancer (disease_9657)	0.576892	0.987054	16	0	0
9	hereditary breast ovarian cancer syndrome (disease_10777)	0.574417	0.994999	12	0	0
10	hereditary breast carcinoma (disease_4152)	0.572119	0.991067	12	0	0

OUTPUT 3 — EVIDENCE & SIMILARITY SUBGRAPH

Drug Repurposing Evidence Graph — psoriasis (model: HGT)
Drugs · Protein Evidence · Phenotypes · Similar Diseases · Safety



Subgraph saved → /mnt/data/demo_subgraph.png

OUTPUT 4 — RAG EXPLANATION & SUMMARY

Disease query : psoriasis
Model : HGT

SIMILAR DISEASES USED AS CONTEXT

Diseases most biologically similar to the query were identified using:
embedding cosine similarity · disease ontology edges · shared proteins
PPI-expanded protein bridges · phenotype overlap · shared therapeutics

- parapsoriasis (disease_14126) (similarity=0.795, embedding=0.985, shared proteins=59, shared effects=0)
- psoriasis 14, pustular (disease_2356) (similarity=0.722, embedding=0.849, shared proteins=57, shared effects=0)
- dermatitis (disease_9036) (similarity=0.666, embedding=0.896, shared proteins=2, shared effects=0)
- lung cancer (disease_16106) (similarity=0.582, embedding=0.955, shared proteins=14, shared effects=0)
- pustular psoriasis (disease_8434) (similarity=0.581, embedding=0.986, shared proteins=1, shared effects=0)

TOP 5 CANDIDATE DRUGS

Rank 1 – Methylprednisolone (drug_929)

Currently known to treat : X-linked adrenal hypoplasia congenita, chondrocalcinosis, dermatitis, atopic (+73 more)
Raw model score : 0.573
Similar disease support : 0.673
Final rerank score : 0.684
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Protein interaction bridge: Methylprednisolone (drug_929) --[drug_carrier_protein]--> ALB (protein_4264) --[protein_interacts_with_protein]--> APOE (protein_7920) <--[disease_associated_w
- Protein interaction bridge: Methylprednisolone (drug_929) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> IL12B (protein_8067) <--[disease_associated
- Protein interaction bridge: Methylprednisolone (drug_929) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> NFKBIA (protein_9739) <--[disease_associate
- Protein interaction bridge: Methylprednisolone (drug_929) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> PPARG (protein_11118) <--[disease_associate

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 5
- Disease-to-disease transfer : 1
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

parapsoriasis (disease_14126) | rheumatoid arthritis (disease_15665)

Rank 2 – Regorafenib (drug_5408)

Currently known to treat : gastrointestinal stromal tumor, pediatric hepatocellular carcinoma, adenocarcinoma of liver and intrahepatic biliary tract (+6 more)
Raw model score : 0.573
Similar disease support : 0.566
Final rerank score : 0.657
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Protein interaction bridge: Regorafenib (drug_5408) --[drug_target_protein]--> EPHA2 (protein_3964) --[protein_interacts_with_protein]--> CYP2S1 (protein_7098) <--[disease_associated_with
- Protein interaction bridge: Regorafenib (drug_5408) --[drug_target_protein]--> EPHA2 (protein_3964) --[protein_interacts_with_protein]--> TP53 (protein_14868) <--[disease_associated_with_
- Protein interaction bridge: Regorafenib (drug_5408) --[drug_target_protein]--> FGFR1 (protein_4564) --[protein_interacts_with_protein]--> STAT3 (protein_14481) <--[disease_associated_with
- Protein interaction bridge: Regorafenib (drug_5408) --[drug_target_protein]--> FGFR2 (protein_4567) --[protein_interacts_with_protein]--> STAT3 (protein_14481) <--[disease_associated_with

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 5
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

hepatocellular carcinoma (disease_14718) | liver cancer (disease_9657)

Rank 3 – Ramucirumab (drug_3456)

Currently known to treat : pediatric hepatocellular carcinoma, undifferentiated carcinoma of liver and intrahepatic biliary tract, liver cancer (+5 more)

Raw model score : 0.573

Similar disease support : 0.851

Final rerank score : 0.628

Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Protein interaction bridge: Ramucirumab (drug_3456) --[drug_target_protein]--> KDR (protein_8358) --[protein_interacts_with_protein]--> STAT3 (protein_14481) <--[disease_associated_with_p
- Protein interaction bridge: Ramucirumab (drug_3456) --[drug_target_protein]--> KDR (protein_8358) --[protein_interacts_with_protein]--> VEGFA (protein_15183) <--[disease_associated_with_p

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 2
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

lung cancer (disease_16106) | hepatocellular carcinoma (disease_14718) | liver cancer (disease_9657)

Rank 4 – Fluocinonide (drug_1014)

Currently known to treat : primary cutaneous T-cell lymphoma, discoid lupus erythematosus, dermatitis (+8 more)

Raw model score : 0.573

Similar disease support : 0.327

Final rerank score : 0.597

Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Protein interaction bridge: Fluocinonide (drug_1014) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> IL12B (protein_8067) <--[disease_associated_with
- Protein interaction bridge: Fluocinonide (drug_1014) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> NFKBIA (protein_9739) <--[disease_associated_wit
- Protein interaction bridge: Fluocinonide (drug_1014) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> PPARG (protein_11118) <--[disease_associated_wit
- Protein interaction bridge: Fluocinonide (drug_1014) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> STAT3 (protein_14481) <--[disease_associated_wit

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 5
- Disease-to-disease transfer : 1
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

dermatitis (disease_9036)

Rank 5 – Doxorubicin (drug_966)

Currently known to treat : acute myeloid leukemia and myelodysplastic syndromes related to topoisomerase type 2 inhibitor, acute myeloid leukemia with t(8;21)(q22;q22) translocation, medullar

Raw model score : 0.573

Similar disease support : 0.285

Final rerank score : 0.586

Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Shared protein target: Doxorubicin (drug_966) --[drug_enzyme_protein]--> NOS2 (protein_9783) <--[disease_associated_with_protein]-- psoriasis (disease_12709)
- Protein interaction bridge: Doxorubicin (drug_966) --[drug_carrier_protein]--> ALB (protein_4264) --[protein_interacts_with_protein]--> APOE (protein_7920) <--[disease_associated_with_protein]--> APOE (protein_7920) <--[disease_associated_with_protein]--> APOE (protein_7920)
- Protein interaction bridge: Doxorubicin (drug_966) --[drug_enzyme_protein]--> NQO1 (protein_3741) --[protein_interacts_with_protein]--> TP53 (protein_14868) <--[disease_associated_with_protein]--> TP53 (protein_14868)
- Protein interaction bridge: Doxorubicin (drug_966) --[drug_enzyme_protein]--> NOS2 (protein_9783) --[protein_interacts_with_protein]--> PPARG (protein_11118) <--[disease_associated_with_protein]--> PPARG (protein_11118)

Evidence profile:

- Shared protein bridges : 1
- Protein interaction bridges : 5
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

lung cancer (disease_16106)

INTERPRETATION NOTE

These are graph-based research hypotheses, not clinical prescriptions. The strongest candidates combine: high model score + strong mechanistic evidence (especially shared protein bridges) + support from biologically similar diseases + no direct contraindication signals.

Always validate predictions against current clinical literature before drawing any experimental or clinical conclusions.

Structured RAG report table:

rank	drug	known_treatment_for	model_score	similar_disease_support	rerank_score	risk	evidence_summary	safety_summary
1	Methylprednisolone (drug_929)	X-linked adrenal hypoplasia congenita, chondrocalcinosis, dermatitis, atopic (+73 more)	0.573084	0.673471	0.683564	low	Protein interaction bridge: Methylprednisolone (drug_929) --[drug_carrier_protein]--> ALB (protein_4264) --[protein_interacts_with_protein]--> APOE (protein_7920) <--[disease_associated_with_protein]--> APOE (protein_7920)	No direct contraindication or adverse overlap signal found.
2	Regorafenib (drug_5408)	gastrointestinal stromal tumor, pediatric hepatocellular carcinoma, adenocarcinoma of liver and intrahepatic biliary...	0.572930	0.565849	0.656573	low	Protein interaction bridge: Regorafenib (drug_5408) --[drug_target_protein]--> EPHA2 (protein_3964) --[protein_interacts_with_protein]--> TP53 (protein_14868) <--[disease_associated_with_protein]--> TP53 (protein_14868)	No direct contraindication or adverse overlap signal found.
3	Ramucirumab (drug_3456)	pediatric hepatocellular carcinoma, undifferentiated carcinoma of liver and intrahepatic biliary tract, liver cancer...	0.572884	0.851147	0.627873	low	Protein interaction bridge: Ramucirumab (drug_3456) --[drug_target_protein]--> KDR (protein_8358) --[protein_interacts_with_protein]--> TP53 (protein_14868) <--[disease_associated_with_protein]--> TP53 (protein_14868)	No direct contraindication or adverse overlap signal found.
4	Fluocinonide (drug_1014)	primary cutaneous T-cell lymphoma, discoid lupus erythematosus, dermatitis (+8 more)	0.572963	0.326529	0.596762	low	Protein interaction bridge: Fluocinonide (drug_1014) --[drug_target_protein]--> NR3C1 (protein_6984) --[protein_interacts_with_protein]--> TP53 (protein_14868) <--[disease_associated_with_protein]--> TP53 (protein_14868)	No direct contraindication or adverse overlap signal found.
5	Doxorubicin (drug_966)	acute myeloid leukemia and myelodysplastic syndromes related to topoisomerase type 2 inhibitor, acute myeloid leukemia...	0.572843	0.285298	0.586388	low	Shared protein target: Doxorubicin (drug_966) --[drug_enzyme_protein]--> NOS2 (protein_9783) <--[disease_associated_with_protein]--> psoriasis (disease_12709)	No direct contraindication or adverse overlap signal found.